# Supplementary Material to "Double Tube-Preserving Regularization for Robust Visual Data Recovery"

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**Abstract**

In the supplementary material, we first recall some necessary preliminaries. Then we present the proofs of Theorem 1 and the convergence analysis of DTR. We also present the optimization algorithm of GDTR. Finally, we provide additional experimental results to further validate the effectiveness of the proposed methods.

## 1. Notations And Preliminaries

Initially, we recall some key notations utilized in this paper. We employ lowercase letters ($t$), bold lowercase letters ($\mathbf{t}$), bold uppercase letters ($\mathbf{T}$), and calligraphic capitals ($\mathcal{T}$) to represent scalars, vectors, matrices, and tensors, respectively. A vector $\mathbf{t} \in \mathbb{R}^d$ is denoted as $\|\mathbf{t}\|_2 = \sqrt{\sum_{i=1}^{d} t_i^2}$. For a tensor $\mathcal{T}$, the $\ell_1$- and Frobenius norms take the forms $\|\mathcal{T}\|_1 = \sum_{ijk} |\mathcal{T}_{ijk}|$ and $\|\mathcal{T}\|_F = \sqrt{\sum_{ijk} \|\mathcal{T}_{ijk}\|^2}$, respectively. In the matrix case, $\|\mathbf{T}\|_* = \sum_l \sigma_l(\mathbf{T})$ represents the nuclear norm, with $\sigma_l(\mathbf{T})$ being the $l$-th singular value. For brevity, $[d]$ represents the integer sequence $\{1, 2, \ldots, d\}$. Key notations and acronyms are summarized in Tab. 1.

**Definition 1.** ***(Tensor Singular Value Decomposition, T-SVD** [1]) For $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$,*

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| Notations/Acronyms | Descriptions | Notations/Acronyms | Descriptions |
|---|---|---|---|
| $\mathcal{T}(i,:,:)$ | $i$-th horizontal slice of $\mathcal{T}$ | $\mathcal{T}(:,j,:)$ | $j$-th lateral slice of $\mathcal{T}$ |
| $\mathcal{T}(:,:,k)$ or $\mathbf{T}^{(k)}$ | $k$-th frontal slice of $\mathcal{T}$ | $\mathcal{T}(i,j,k)$ or $\mathcal{T}_{ijk}$ | $(i,j,k)$-th entry of $\mathcal{T}$ |
| $\mathcal{T}^*$ | transpose of $\mathcal{T}$ | $\mathcal{T}_1 * \mathcal{T}_2$ | tensor product of $\mathcal{T}_1$ and $\mathcal{T}_2$ |
| DFT | Discrete Fourier Transform | $\bar{\mathcal{T}} = fft(\mathcal{T},[],3)$ | DFT of $\mathcal{T}$ along the 3-rd dimension |
| $\vec{\sigma}_l(\bar{\mathcal{L}})$ | $l$-th singular tube of $\mathcal{L}$ | $\mathcal{E}(i,j,:)$ | $(i,j)$-th noise tube of $\mathcal{E}$ |
| $\|\mathcal{T}\|_*$ | Tensor Nuclear Norm | $\|\mathcal{T}\|_1$ | Tensor $\ell_1$ Norm |
| $[n]$ | index set $\{1,2,\cdots,n\}$ | TNN | Tensor Nuclear Norm |
| T2NN | Tube-preserving TNN | T2L1 | Tube-preserving Tensor $\ell_1$ Norm |

Table 1: Key acronyms and notations utilized in this work.

*its t-SVD is expressed as*

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*, \tag{1}$$

*where* $\mathcal{U} \in \mathbb{R}^{d_1\times d_1\times d_3}$, $\mathcal{V} \in \mathbb{R}^{d_2\times d_2\times d_3}$ *are orthogonal, i.e.,* $\mathcal{U}*\mathcal{U}^* = \mathcal{U}^**\mathcal{U} = \mathcal{V}*\mathcal{V}^* = \mathcal{V}^**\mathcal{V} = \mathcal{I}$. *The tensor* $\mathcal{S} \in \mathbb{R}^{d_1\times d_2\times d_3}$ *is f-diagonal, meaning that all its frontal slices are diagonal matrices. The singular value tensor is defined by* $\bar{\mathcal{S}} = fft(\mathcal{S},[],3)$.

**Definition 2.** *(**Tensor tubal rank** [1]) Given* $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$, *its tubal rank* $rank_t(\mathcal{X})$ *is determined by the number of nonzero singular tubes of in* $\mathcal{S}$ *obtained from the T-SVD of* $\mathcal{X}$. *Specifically,* $rank_t(\mathcal{X}) = \#\{l, \mathcal{S}(l,l,:) \neq \mathbf{0}\}$.

**Definition 3.** *(**Tensor Nuclear Norm** [1]) Given* $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$ *and* $d = \min(d_1,d_2)$, *the tensor nuclear norm (TNN) is given by*

$$\|\mathcal{X}\|_* = \frac{1}{d_3}\sum_{k=1}^{d_3}\|\bar{\mathbf{X}}^{(k)}\|_* = \frac{1}{d_3}\sum_{k=1}^{d_3}\sum_{l=1}^{d}\sigma_l\left(\bar{\mathbf{X}}^{(k)}\right). \tag{2}$$

## 2. Proposed Method

### 2.1. Proof of Theorem 1

**Theorem 1.** *Given* $\tau \geq 0$ *and a weight vector* $\mathbf{w} \in \mathbb{R}^d$, *with* $d = min(d_1,d_2)$, *if* $\mathbf{w}$ *satisfies* $0 \leq w_1 \leq \cdots \leq w_d$, *the T2NN proximal operator (6) of the tensor* $\mathcal{M} \in \mathbb{R}^{d_1\times d_2\times d_3}$ *obeys*

$$\mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}) = \underset{\mathcal{L}\in\mathbb{R}^{d_1\times d_2\times d_3}}{\arg\min}\ \frac{1}{2}\|\mathcal{L}-\mathcal{M}\|_F^2 + \tau\|\mathcal{L}\|_{T,*}. \tag{3}$$

The proof is based on the matrix Weighted Nuclear Norm Minimization (WNNM) problem. We first recall the following result, which will be used to derive the proximal operator of T2NN.

**Lemma 1.** *[2] For* $\mathbf{M} \in \mathbb{R}^{d_1 \times d_2}$ *with SVD* $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ *and a weight vector* $\mathbf{w} = [w_1, \cdots, w_d]^T \in \mathbb{R}^d$ *with* $d = \min(d_1, d_2)$*. We now tackle the WNNM problem*

$$\mathbf{prox}_{\|\cdot\|_{\mathbf{w},*}}(\mathbf{M}) := \arg\min_{\mathbf{L} \in \mathbb{R}^{d_1 \times d_2}} \frac{1}{2}\|\mathbf{L} - \mathbf{M}\|_F^2 + \|\mathbf{L}\|_{\mathbf{w},*}, \tag{4}$$

*where* $\mathbf{prox}_{\|\cdot\|_{\mathbf{w},*}}(\mathbf{M})$ *represents the proximal mapping associated with the weighted nuclear norm. Under the condition* $0 \leq w_1 \leq \cdots \leq w_d$*, the minimizer of Eq. (4) is obtained as*

$$\mathbf{L}^* = \mathbf{prox}_{\|\cdot\|_{\mathbf{w},*}}(\mathbf{M}) = \mathbf{U}\mathbf{P}_{\mathbf{w}}(\mathbf{\Sigma})\mathbf{V}^T.$$

*Here,* $\mathbf{P}_{\mathbf{w}}(\mathbf{\Sigma})$ *denotes a diagonal matrix obtained by applying the shrinkage operation to* $\mathbf{\Sigma}$*, with its l-th diagonal entry given by* $(\mathbf{P}_{\mathbf{w}}(\mathbf{\Sigma}))_{ll} = (\mathbf{\Sigma}_{ll} - w_l)_+$*. The operator* $(\cdot)_+$ *represents the nonnegative truncation, i.e.,* $(z)_+ = \max(z, 0)$*.*

Now we consider the T2NN minimization problem

$$\arg\min_{\mathcal{L} \in \mathbb{R}^{d_1 \times d_2 \times d_3}} \frac{1}{2}\|\mathcal{L} - \mathcal{M}\|_F^2 + \tau\|\mathcal{L}\|_{T,*}. \tag{5}$$

We can give an optimal solution of the problem (5) according to Lemma 1.

For $\mathcal{M} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ and the vector $\mathbf{w} = [w_1, \cdots, w_d] \in \mathbb{R}^d$ with $d = min(d_1, d_2)$, $\mathcal{M} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$ denote its T-SVD. For $\tau \geq 0$, the T2NN proximal mapping can be written as

$$\mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}) = \mathcal{U} * ifft\left(\mathcal{P}_{\tau\mathbf{w}}(\bar{\mathcal{S}}), [], 3\right) * \mathcal{V}^*. \tag{6}$$

For $k \in [d_3]$, we compute the SVD of $\bar{\mathbf{M}}^{(k)}$ to obtain the corresponding singular value matrix $\bar{\mathbf{S}}^{(k)}$. The tensor $\mathcal{P}_{\tau\mathbf{w}}(\bar{\mathcal{S}})$ is then constructed by applying $\mathbf{P}_{\tau\mathbf{w}}(\cdot)$ entry-wise to each frontal slice, i.e., its $k$-th slice equals $\mathbf{P}_{\tau\mathbf{w}}\left(\bar{\mathbf{S}}^{(k)}\right)$.

With these preparations, we proceed to prove Theorem 1.

*Proof.* By using the property $\|\bar{\mathcal{L}}\|_F^2 = \frac{1}{d_3}\|\mathcal{L}\|_F^2$, the problem (5) is equivalent to

$$\arg\min_{\mathcal{L} \in \mathbb{R}^{d_1 \times d_2 \times d_3}} \frac{1}{2d_3}\left\|\bar{\mathcal{L}} - \bar{\mathcal{M}}\right\|_F^2 + \frac{\tau}{d_3}\sum_{k=1}^{d_3}\left\|\bar{\mathbf{L}}^{(k)}\right\|_{\mathbf{w},*}, \tag{7}$$

where $\mathbf{w} = (w_1, \cdots, w_d)$, $p_l = f_{\mathcal{L}}\left(\left\|\vec{\sigma}_l(\bar{\mathcal{X}})\right\|_2\right)$ denotes the vector of function values for singular tubes. The problem can be decomposed into $d_3$ independent subproblems

$$\arg\min_{\mathcal{L}\in\mathbb{R}^{d_1\times d_2\times d_3}} \frac{1}{2}\left\|\bar{\mathbf{L}}^{(k)} - \bar{\mathbf{M}}^{(k)}\right\|_F^2 + \tau\left\|\bar{\mathbf{L}}^{(k)}\right\|_{\mathbf{w},*}, \tag{8}$$

for $k \in [d_3]$. Consider the SVD of $\bar{\mathbf{M}}^{(k)}$ as $\bar{\mathbf{M}}^{(k)} = \bar{\mathbf{U}}^{(k)}\bar{\mathbf{S}}^{(k)}\left(\bar{\mathbf{V}}^{(k)}\right)^T$. According to Lemma 1, the solution of (8) can be expressed as

$$\bar{\mathbf{L}}^{(k)} = \bar{\mathbf{U}}^{(k)}\mathbf{P}_{\tau\mathbf{w}}\left(\bar{\mathbf{S}}^{(k)}\right)\left(\bar{\mathbf{V}}^{(k)}\right)^T.$$

Let $\mathcal{U} = ifft(\bar{\mathcal{U}}, [], 3)$ and $\mathcal{V} = ifft(\bar{\mathcal{V}}, [], 3)$. By leveraging the properties of the t-product, we derive the optimal solution to (5) as

$$\mathcal{L}^* = \mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}) = \mathcal{U} * ifft\left(\mathcal{P}_{\tau\mathbf{w}}(\bar{\mathcal{S}}), [], 3\right) * \mathcal{V}^*.$$

This completes the proof. □

### *2.2. Convergence Analysis*

This part aims to establishing the convergence analysis for Algorithm 1 in the main paper. Note that different from previous results which employ the fixed and predefined variables, both of the variables $\mathbf{w}$ and $\mathbf{W}$ are adaptively learned in the iterations, which makes the analysis more challenging.

**Lemma 2.** *The sequences* $\{\mathcal{N}_t\}_{t=1}^{\infty}$, $\{\mathbf{w}_t\}_{t=1}^{\infty}$, $\{\mathbf{W}_t\}_{t=1}^{\infty}$ *generated by Algorithm 1 are bounded.*

*Proof.* According to Eqs. (17) and (18) in the main paper, each entry of $\mathbf{w}_t$ and $\mathbf{W}_t$ is bounded by 1. In light of the rule of the Lagrangian multiplier $\mathcal{N}$, we have

$$\begin{aligned}
\|\mathcal{N}_{t+1}\|_F &= \|\mathcal{N}_t + \mu_t\left(\mathcal{L}_{t+1} + \mathcal{E}_{t+1} - \mathcal{X}\right)\|_F \\
&= \mu_t\left\|\mathcal{E}_{t+1} - \left(\mathcal{X} - \mathcal{L}_{t+1} - \mathcal{N}_t/\mu_t\right)\right\|_F \\
&= \mu_t\left\|\mathcal{E}_{t+1} - \mathcal{K}_t\right\|_F \\
&= \mu_t\left\|Q\left(\mathcal{K}_t, \frac{\lambda}{\mu_t}\mathbf{W}_t\right) - \mathcal{K}_t\right\|_F \\
&\le \sum_{i=1}^{d_1}\sum_{j=1}^{d_2}\sum_{k=1}^{d_3}\mu_t\frac{\lambda}{\mu_t}|(\mathbf{W}_t)_{ij}| \\
&\le \lambda d_1 d_2 d_3,
\end{aligned} \tag{9}$$

where the last inequation utilizes the fact that $|(\mathbf{W}_t)_{ij}| = 1/(1 + \mathbf{S}_{t+1}(i,j)/\tau_{\mathcal{E}}) \le 1$. □

**Lemma 3.** *The sequences* $\{\mathcal{L}_t\}_{t=1}^{\infty}$, $\{\mathcal{E}_t\}_{t=1}^{\infty}$ *generated by Algorithm 1 are bounded.*

*Proof.* For simplicity, let $\Theta_t = \{\mathcal{L}_t, \mathcal{E}_t\}$ for $t = 1, \cdots, \infty$.The following proof consists of three steps

- **Step 1:** The sequence $\{L(\Theta_{t+1}, \mathcal{N}_t, \mu_t)\}_{t=1}^{\infty}$ is bounded.
- **Step 2:** The sequences $\{\|\mathcal{L}_t\|_{T,*}\}_{t=1}^{\infty}$ and $\{\|\mathcal{E}_t\|_{T,1}\}_{t=1}^{\infty}$ are bounded.
- **Step 3:** The sequences $\{\mathcal{L}_t\}_{t=1}^{\infty}$ and $\{\mathcal{E}_t\}_{t=1}^{\infty}$ are bounded.

**Proof of Step 1:** Since $\mathcal{L}_{t+1}, \mathcal{E}_{t+1}$ are optimal solutions to the corresponding sub-problems with fixed $\mathbf{w}_t$ and $\mathbf{W}_t$, we have

$$L(\Theta_{t+1}, \mathcal{N}_t, \mu_t) \leq L(\Theta_t, \mathcal{N}_t, \mu_t). \tag{10}$$

Besides, based on the Lagrangian function in Eq. (11) of the main paper, we have

$$\begin{aligned} L(\Theta_t, \mathcal{N}_t, \mu_t) =& L(\Theta_t, \mathcal{N}_{t-1}, \mu_{t-1}) + \frac{\mu_t - \mu_{t-1}}{2}\|\mathcal{L}_t + \mathcal{E}_t - \mathcal{X}\|_F^2 \\ &+ < \mathcal{L}_t + \mathcal{E}_t - \mathcal{X}, \mathcal{N}_t - \mathcal{N}_{t-1} > . \end{aligned} \tag{11}$$

According to the update rule of the Lagrangian multiplier $\mathcal{N}_t = \mathcal{N}_{t-1} + \mu_{t-1}(\mathcal{L}_t + \mathcal{E}_t - \mathcal{X})$, we have

$$\mathcal{L}_t + \mathcal{E}_t - \mathcal{X} = \frac{1}{\mu_{t-1}}(\mathcal{N}_t - \mathcal{N}_{t-1}). \tag{12}$$

Applying it into Eq. (11), we have

$$L(\Theta_t, \mathcal{N}_t, \mu_t) \leq L(\Theta_t, \mathcal{N}_{t-1}, \mu_{t-1}) + \frac{\mu_{t-1} + \mu_t}{2\mu_{t-1}^2}\|\mathcal{N}_t - \mathcal{N}_{t-1}\|_F^2. \tag{13}$$

According to Lemma 2, the sequence $\{\|\mathcal{N}_t\|_F^2, t = 1, \cdots, \infty\}$ is upper bounded. Let $C$ be the upper bound of $\{\|\mathcal{N}_t - \mathcal{N}_{t-1}\|_F^2, t = 1, \cdots, \infty\}$. Combing Eqs. (10) and (13), we have

$$L(\Theta_{t+1}, \mathcal{N}_t, \mu_t) \leq L(\Theta_1, \mathcal{N}_0, \mu_0) + C\sum_{t=1}^{\infty}\frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2}. \tag{14}$$

According to [3], we have that the penalty parameter $\{\mu_t\}$ satisfies $\sum_{t=1}^{\infty}\frac{\mu_{t+1}}{\mu_t^2} < +\infty$, then,

$$\sum_{t=1}^{\infty}\frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2} \leq \sum_{t=1}^{\infty}\frac{\mu_t}{\mu_{t-1}^2} < +\infty. \tag{15}$$

Thus, the sequence $\{L(\Theta_{t+1}, \mathcal{N}_t, \mu_t)\}_{t=1}^{\infty}$ is upper bounded.

**Proof of Step 2:** For convenience, let $\Theta_t = \{\mathcal{L}_t, \mathcal{E}_t\}$ for $t = 1, \cdots, \infty$. We now consider

$$\begin{aligned}
&\|\mathcal{L}_t\|_{T,*} + \|\mathcal{E}_t\|_{T,1} \\
&= L(\Theta_t, \mathcal{N}_{t-1}, \mu_{t-1}) - \frac{\mu_{t-1}}{2}\|\mathcal{L}_t + \mathcal{E}_t - \mathcal{X} + \frac{\mathcal{N}_{t-1}}{\mu_{t-1}}\|_F^2 \\
&+ \frac{1}{2\mu_{t-1}}\|\mathcal{N}_{t-1}\|_F^2 \\
&= L(\Theta_t, \mathcal{N}_{t-1}, \mu_{t-1}) - \frac{\mu_{t-1}}{2}\|\frac{\mathcal{N}_t - \mathcal{N}_{t-1}}{\mu_{t-1}} + \frac{\mathcal{N}_{t-1}}{\mu_{t-1}}\|_F^2 \\
&+ \frac{1}{2\mu_{t-1}}\|\mathcal{N}_{t-1}\|_F^2 \\
&= L(\Theta_t, \mathcal{N}_{t-1}, \mu_{t-1}) - \frac{1}{2\mu_{t-1}}\|\mathcal{N}_t\|_F^2 + \frac{1}{2\mu_{t-1}}\|\mathcal{N}_{t-1}\|_F^2.
\end{aligned} \tag{16}$$

Hence, both $\{\|\mathcal{L}_t\|_{T,*}\}_{t=1}^{\infty}$ and $\{\|\mathcal{E}_t\|_{T,1}\}_{t=1}^{\infty}$ are upper bounded.

**Proof of Step 3:** We now turn to proving that sequences $\{\mathcal{L}_t\}_{t=1}^{\infty}$ and $\{\mathcal{E}_t\}_{t=1}^{\infty}$ are bounded. We first prove the boundness of $\{\mathcal{E}_t\}_{t=1}^{\infty}$ by contradiction and then the proof for $\{\mathcal{L}_t\}_{t=1}^{\infty}$ is analogous. Assume that $\{\mathcal{E}_t\}_{t=1}^{\infty}$ is unbounded. Let $e_t = \max_{ijk}|(\mathcal{E}_t)_{ijk}|$ and then $\{e_t\}_{t=1}^{\infty}$ is also unbounded. For simplicity, let $\overline{S_{ij}} = \text{mean}(S_{ij}, (i,j) \in [d_1] \times [d_2])$. Recalling the definition of $\|\cdot\|_{T,1}$, we obtain

$$\begin{aligned}
\|\mathcal{E}_{t+1}\|_{T,1} &= \sum_{i=1}^{d_1}\sum_{j=1}^{d_2}\sum_{k=1}^{d_3}(\mathbf{W}_t)_{ij}|(\mathcal{E}_t)_{ijk}| \\
&= \sum_{i=1}^{d_1}\sum_{j=1}^{d_2}\sum_{k=1}^{d_3}\frac{1}{1+\frac{S_{ij}}{\tau_{\mathcal{E}}}}|(\mathcal{E}_t)_{ijk}| \\
&\geq \max_{ijk}|(\mathcal{E}_t)_{ijk}| \cdot 1/\left(1 + \frac{\sum_{i=1}^{d_1}\sum_{j=1}^{d_2} S_{ij}}{\gamma_{\mathcal{E}} * \overline{S_{ij}}}\right) \\
&= e_t \cdot 1/(1 + \frac{d_1 d_2}{\gamma_{\mathcal{E}}}).
\end{aligned} \tag{17}$$

Since $1/(1 + \frac{d_1 d_2}{\gamma_{\mathcal{E}}})$ is a fixed positive value and $\{e_t\}_{t=1}^{\infty}$ is unbounded, $\{\|\mathcal{E}_t\|_{T,1}\}_{t=1}^{\infty}$ is also unbounded, which contradicts the results in Step 2. Therefore, $\{\mathcal{E}_t\}_{t=1}^{\infty}$ is bounded. Analogously, $\{\mathcal{L}_t\}_{t=1}^{\infty}$ is also bounded.

This completes the proof. □

With the aforementioned results, we are now ready to prove that the proposed al-

gorithm is convergent.

**Theorem 2.** *For the sequences* $\{\mathcal{L}_t\}_{t=1}^{\infty}$, $\{\mathcal{E}_t\}_{t=1}^{\infty}$ *generated by Algorithm 1, we have*

*(1)* $\lim_{t\to\infty} \|\mathcal{L}_t + \mathcal{E}_t - \mathcal{X}\|_F = 0$.

*(2)* $\lim_{t\to\infty} \|\mathcal{L}_{t+1} - \mathcal{L}_t\|_F = 0$.

*(3)* $\lim_{t\to\infty} \|\mathcal{E}_{t+1} - \mathcal{E}_t\|_F = 0$.

*Proof.* We have proved above that $\{\mathcal{L}_t\}_{t=1}^{\infty}$, $\{\mathcal{E}_t\}_{t=1}^{\infty}$ and $\{\mathcal{N}_t\}_{t=1}^{\infty}$ are all bounded. Hence, there exists at least one accumulation point for $\{\mathcal{L}_t, \mathcal{E}_t, \mathcal{N}_t\}_{t=1}^{\infty}$. Thus, we have

$$\lim_{t\to\infty} \|\mathcal{L}_t + \mathcal{E}_t - \mathcal{X}\|_F = \lim_{t\to\infty} \frac{1}{\mu_{t-1}} \|\mathcal{N}_t - \mathcal{N}_{t-1}\|_F = 0, \tag{18}$$

and the accumulation point satisfies the feasibility condition of the optimization problem.

By Eq. (12), we have that $\mathcal{L}_t = \mathcal{X} - \mathcal{E}_t + (\mathcal{N}_t - \mathcal{N}_{t-1})/\mu_{t-1}$, and recall $\mathcal{L}_{t+1} = \mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}_t)$ and $\mathcal{M}_t = \mathcal{X} - \mathcal{E}_t - \mathcal{N}_t/\mu_t$. Hence, there holds that

$$\begin{aligned}
&\|\mathcal{L}_{t+1} - \mathcal{L}_t\|_F \\
&\le \left\|\mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}_t) - \mathcal{M}_t\right\|_F + \|\mathcal{M}_t - \mathcal{L}_t\|_F \\
&= \frac{1}{\sqrt{d_3}} \left\|\mathrm{bdiag}(\overline{\mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}_t)}) - \mathrm{bdiag}(\overline{\mathcal{M}_t})\right\|_F \\
&+ \|(\mathcal{N}_t - \mathcal{N}_{t-1})/\mu_{t-1} + \mathcal{N}_t/\mu_t\|_F \\
&\le \frac{1}{\sqrt{d_3}} \sqrt{\sum_{k=1}^{d_3}\sum_{l=1}^{d} (\frac{(w_t)_l}{\mu_t})^2} + \frac{\|\mathcal{N}_t - \mathcal{N}_{t-1}\|_F}{\mu_{t-1}} + \frac{\|\mathcal{N}_t\|_F}{\mu_t} \\
&= \frac{\|\mathbf{w}_t\|_2}{\mu_t} + \frac{\|\mathcal{N}_t - \mathcal{N}_{t-1}\|_F}{\mu_{t-1}} + \frac{\|\mathcal{N}_t\|_F}{\mu_t}.
\end{aligned} \tag{19}$$

Since $\lim_{t\to\infty} \frac{1}{\mu_t} = 0$, and the sequences $\{\mathbf{w}_t\}_{t=1}^{\infty}$ and $\{\mathcal{N}_t\}_{t=1}^{\infty}$ are bounded, we have

$$\lim_{t\to\infty} \|\mathcal{L}_{t+1} - \mathcal{L}_t\|_F = 0. \tag{20}$$

Analogously, by Eq. (12) we have that $\mathcal{E}_t = \mathcal{X} - \mathcal{L}_t + (\mathcal{N}_t - \mathcal{N}_{t-1})/\mu_{t-1}$, and recall

$\mathcal{K}_t = \mathcal{X} - \mathcal{L}_{t+1} - \mathcal{N}_t/\mu_t$ and $\mathcal{E}_{t+1} = Q\left(\mathcal{K}_t, \frac{\lambda}{\mu_t}\mathbf{W}_t\right)$. It follows that

$$\begin{aligned}
&\|\mathcal{E}_{t+1} - \mathcal{E}_t\|_F \\
&= \left\| Q\left(\mathcal{K}_t, \frac{\lambda}{\mu_t}\mathbf{W}_t\right) - \mathcal{K}_t + \mathcal{L}_t - \mathcal{L}_{t+1} - \frac{\mathcal{N}_t}{\mu_t} - \frac{\mathcal{N}_t - \mathcal{N}_{t-1}}{\mu_{t-1}} \right\|_F \\
&\leq \left\| Q\left(\mathcal{K}_t, \frac{\lambda}{\mu_t}\mathbf{W}_t\right) - \mathcal{K}_t \right\|_F + \|\mathcal{L}_t - \mathcal{L}_{t+1}\|_F \\
&+ \left\| \frac{\mathcal{N}_t}{\mu_t} + \frac{\mathcal{N}_t - \mathcal{N}_{t-1}}{\mu_{t-1}} \right\|_F \\
&\leq \frac{\lambda}{\mu_t} d_1 d_2 d_3 + \|\mathcal{L}_t - \mathcal{L}_{t+1}\|_F + \frac{\|\mathcal{N}_t - \mathcal{N}_{t-1}\|_F}{\mu_{t-1}} + \frac{\|\mathcal{N}_t\|_F}{\mu_t}.
\end{aligned} \tag{21}$$

Considering Eq. (20) and $\lim_{t\to\infty} \frac{1}{\mu_t} = 0$, there holds

$$\lim_{t\to\infty} \|\mathcal{E}_{t+1} - \mathcal{E}_t\|_F = 0. \tag{22}$$

This completes the proof. □

### *2.3. Optimization of GDTR*

This subsection describes the optimization procedure for Tensor Robust Principal Component Analysis via Gradient Double Tube-preserving Regularization (GDTR).

Recall that the GDTR model can be reformulated as

$$\begin{aligned}
&\min_{\mathcal{L},\mathcal{G}_m,\mathcal{E}} \frac{1}{|\Gamma|} \sum_{m\in\Gamma} \|\mathcal{G}_m\|_{T,*} + \lambda\|\mathcal{E}\|_{T,1} \\
&\text{s.t. } \mathcal{G}_m = \nabla_m(\mathcal{L}), \mathcal{X} = \mathcal{L} + \mathcal{E}.
\end{aligned} \tag{23}$$

Then the augmented Lagrangian function of (23) is

$$\begin{aligned}
&L(\mathcal{L}, \{\mathcal{G}_m, m \in \Gamma\}, \mathcal{E}, \{\Lambda_m, m \in \Gamma\}, \mathcal{R}) = \\
&\sum_{m\in\Gamma} \left(\frac{1}{|\Gamma|}\|\mathcal{G}_m\|_{T,*} + < \Lambda_m, \nabla_m(\mathcal{L}) - \mathcal{G}_m > + \frac{\mu}{2}\|\nabla_m(\mathcal{L}) - \mathcal{G}_m\|_F^2\right) \\
&+ \lambda\|\mathcal{E}\|_{T,1} + < \mathcal{R}, \mathcal{X} - \mathcal{L} - \mathcal{E} > + \frac{\mu}{2}\|\mathcal{X} - \mathcal{L} - \mathcal{E}\|_F^2,
\end{aligned} \tag{24}$$

where $\mu$ is a positive scalar, and $\Lambda_m$ and $\mathcal{R}$ are the Lagrangian multipliers. The above formulation can be equivalently rewritten as

$$\begin{aligned}
L = &\sum_{m\in\Gamma} \left(\frac{1}{|\Gamma|}\|\mathcal{G}_m\|_{T,*} + \frac{\mu}{2}\|\nabla_m(\mathcal{L}) - \mathcal{G}_m + \Lambda_m/\mu\|_F^2\right) \\
&+ \lambda\|\mathcal{E}\|_{T,1} + \frac{\mu}{2}\|\mathcal{X} - \mathcal{L} - \mathcal{E} + \mathcal{R}/\mu\|_F^2 + C,
\end{aligned} \tag{25}$$

where $C$ contains only the squared terms that depend on the multipliers.

To facilitate the derivation of the update steps, we define the adaptive weights of the structure functions as

$$w_l = f_{\mathcal{L}}\left(\left\|\vec{\sigma}_l(\bar{\mathcal{G}}_m)\right\|_2\right), \quad W_{ij} = f_{\mathcal{E}}\left(\|\mathcal{E}(i,j,:)\|_2\right),$$

where $l \in [d]$, $d = \min(d_1, d_2)$ for $w_l$, and $i \in [d_1]$, $j \in [d_2]$ for $W_{ij}$. Here, $w_l$ denotes the adaptive weight for the $l$-th singular value tube of $\bar{\mathcal{G}}_m$, and $W_{ij}$ denotes the adaptive weight for the $(i,j)$-th tube of $\mathcal{E}$. These values are assembled into a weight vector $\mathbf{w}_m = [w_1, \cdots, w_d] \in \mathbb{R}^d$ and a weight matrix $\mathbf{W} \in \mathbb{R}^{d_1 \times d_2}$ with $(i,j)$-th entry $W_{ij}$.

Next, we derive the solutions of the subproblems associated with each variable.

**Update $\mathcal{L}$:** To update $\mathcal{L}$ (fixing other variables), we take the derivative in (25) with respect to $\mathcal{L}$, and then we get the following linear system

$$(\mathcal{I} + \sum_{m\in\Gamma} \nabla_m^T \nabla_m)(\mathcal{L}) = \mathcal{X} - \mathcal{E} + \mathcal{R}/\mu + \sum_{m\in\Gamma} \nabla_m^T(\mathcal{G}_m - \Lambda_m/\mu), \tag{26}$$

where $\nabla_m^T(\cdot)$ denotes the transpose operator of $\nabla_m(\cdot)$. Since the tensor difference operator is linear under the tensor product formulation [4], the multi-dimensional Fourier transform $\mathcal{F}(\cdot)$ can be employed to convert the corresponding difference tensors $\mathcal{D}_m$ into diagonal forms. Consequently, the solution of (26) can be efficiently obtained by exploiting the properties of the Fourier transform, i.e.,

$$\mathcal{L}^{t+1} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\mathcal{X} - \mathcal{E}^t + \mathcal{R}^t/\mu_t) + \mathcal{H}}{\mathbf{1} + \sum_{m\in\Gamma} \mathcal{F}(\mathcal{D}_m)^* \odot \mathcal{F}(\mathcal{D}_m)}\right), \tag{27}$$

where $\mathcal{H} = \sum_{m\in\Gamma} \mathcal{F}(\mathcal{D}_m)^* \odot \mathcal{F}(\mathcal{G}_m^t - \Lambda_m^t/\mu_t)$, $\mathbf{1}$ is a tensor with all entries as 1, $\odot$ represents the element-wise multiplication operator, and the division operation is also performed element by element.

**Update $\mathcal{G}_m, m \in \Gamma$:** For each $m \in \Gamma$, fixing other variables in (25), we can get

$$\mathcal{G}_m^{t+1} = \arg\min_{\mathcal{G}_m} \frac{1}{|\Gamma|}\|\mathcal{G}_m\|_{T,*} + \frac{\mu_t}{2}\|\nabla_m(\mathcal{L}^{t+1}) - \mathcal{G}_m + \Lambda_m^t/\mu_t\|_F^2. \tag{28}$$

According to Theorem 1, we obtain the solution of this subproblem as

$$\mathcal{G}_m^{t+1} = \mathbf{prox}_{\frac{1}{|\Gamma|\mu_t}\|\cdot\|_{T,*}}(\nabla_m(\mathcal{L}^{t+1}) + \Lambda_m^t/\mu_t), \tag{29}$$

where the proximal operator is computed with weight vector $\mathbf{w}_m$ corresponding to the singular tubes of $\bar{\mathcal{G}}_m$.

**Update $\mathcal{E}$:** We update $\mathcal{E}$ by solving the subproblem below:

$$\mathcal{E}^{t+1} = \underset{\mathcal{E}\in\mathbb{R}^{d_1\times d_2\times d_3}}{\arg\min} \frac{1}{2}\left\|\mathcal{E} - \left(\mathcal{X} - \mathcal{L}^{t+1} + \mathcal{R}^t/\mu_t\right)\right\|_F^2 + \frac{\lambda}{\mu_t}\|\mathcal{E}\|_{T,1}. \tag{30}$$

To solve the subproblem above, we now define a tensor soft-thresholding operator $\mathcal{Q}$ by making that $(\mathcal{Q}(\mathcal{K}, \mathbf{W}))_{ijk} = \text{sign}(\mathcal{K}_{ijk})(|\mathcal{K}_{ijk}| - \mathbf{W}_{ij})_+$ where $(z)_+ = z$ if $z > 0$ and $(z)_+ = 0$ if $z \leq 0$. Then we can get the solution of Eq. (30) as

$$\mathcal{E}^{t+1} = \mathcal{Q}\left(\mathcal{K}^t, \frac{\lambda}{\mu_t}\mathbf{W}^t\right), \tag{31}$$

where $\mathcal{K}^t = \mathcal{X} - \mathcal{L}^{t+1} + \mathcal{R}^t/\mu_t$.

**Update $\mathbf{w}_m, m \in \Gamma$ and $\mathbf{W}$:** By the definitions of T2NN and T2L1 in the main paper, we can update $\mathbf{w}_m$ and $\mathbf{W}$ as follows.

We first update the weight vector $\mathbf{w}_m$ for singular tubesof each $\bar{\mathcal{G}}_m$ by

$$w_l = \frac{1}{1 + \left\|\vec{\sigma}_l(\bar{\mathcal{G}}_{m,t+1})\right\|_2/\tau_{\mathcal{L}}}, \ l \in [d], \tag{32}$$

where $[d] = \{1, 2, \ldots, d\}$. We then update the weight matrix $\mathbf{W}$ of $\mathcal{E}$ by

$$W_{ij} = \frac{1}{1 + \|\mathcal{E}_{t+1}(i, j, :)\|_2/\tau_{\mathcal{E}}}, \ i \in [d_1], \ j \in [d_2], \tag{33}$$

where $\tau_{\mathcal{L}}$ and $\tau_{\mathcal{E}}$ are positive parameters.

**Update $\Lambda_m$ and $\mathcal{R}$:** We can update $\Lambda_m$ and $\mathcal{R}$ by

$$\Lambda_m^{t+1} = \Lambda_m^t + \mu_t(\nabla_m(\mathcal{L}^{t+1}) - \mathcal{G}_m^{t+1}), \forall m \in \Gamma, \tag{34}$$

$$\mathcal{R}^{t+1} = \mathcal{R}^t + \mu_t(\mathcal{X} - \mathcal{L}^{t+1} - \mathcal{E}^{t+1}). \tag{35}$$

Finally, $\mu$ is updated by $\mu_{t+1} = \rho\mu_t$, where $\rho = 1.1$. Algorithm 21 provides the pseudo-code of GDTR.

## 3. Additional Experimental Results

Additional experiments are conducted to further assess the performance of DTR and GDTR.

**Algorithm 2** TRPCA via Gradient Double Tube-preserving Regularization (GDTR)

**Input:** Observation $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, priori set $\Gamma$ and the parameter $\lambda$.

**Initialize:** $\mathcal{L}^0 = \mathbf{0}, \mathcal{E}^0 = \mathbf{0}, \mathcal{R}^0 = \mathbf{0}, \mathcal{G}_m^0 = \mathbf{0}, \Lambda_m^0 = \mathbf{0}, \mathbf{w}_m^0 = \mathbf{1}_d, \mathbf{W}^0 = \mathbf{1}_{d_1 \times d_2}, \mu_0 = 10^{-2}$, $\rho = 1.1$, and $t = 0$.

**while** not converged **do**

1: Update $\mathcal{L}^{t+1}$ by Eq. (27).

2: Update $\mathcal{G}_m^{t+1}$ by Eq. (29) for each $m \in \Gamma$.

3: Update $\mathcal{E}^{t+1}$ by Eq. (30).

3: Update $\mathbf{w}_m^{t+1}$ for each $m \in \Gamma$ and $\mathbf{W}^{t+1}$ by Eqs. (32) and (33), respectively.

4: Update multipliers $\Lambda_m^{t+1}, m \in \Gamma$ and $\mathcal{R}^{t+1}$ by Eqs. (34) and (35), respectively.

5: Let $\mu_{t+1}$ by $\mu_{t+1} = \rho\mu_t;\ t = t + 1$.

**end while**

**Output:** $\mathcal{L} = \mathcal{L}_{t+1}, \mathcal{E} = \mathcal{E}_{t+1}$.

**Datasets:** The Berkeley Segmentation Dataset (BSD) [5] is used as benchmark. It contains 500 color images. Following [1], we corrupt each color image by randomly replacing 10% of its pixels with arbitrary values drawn from [0, 255]. The corruptions are on the whole tubes and all 3 channels are corrupted at the same positions.

**Comparison methods:** The proposed methods are evaluated in comparison with several representative methods, including RPCA [6], SNN [7], TRPCA [1], ETRPCA [8] and t-CTV [4]. For DTR and GDTR, we set $\lambda = 1/\sqrt{3\max(d_1, d_2)}$ recommended in theory [4]. We use the parameters recommended by the authors. We adopt the Peak Signal-to- Noise Ratio (PSNR) to quantify the quality of recovery images [9].

**Results:** Fig. 1 presents the PSNR results of several methods over the first 50 BSD images. The above observations can be summarized as follows:

- Firstly, the recovery performance of DTR is more outstanding than TRPCA,

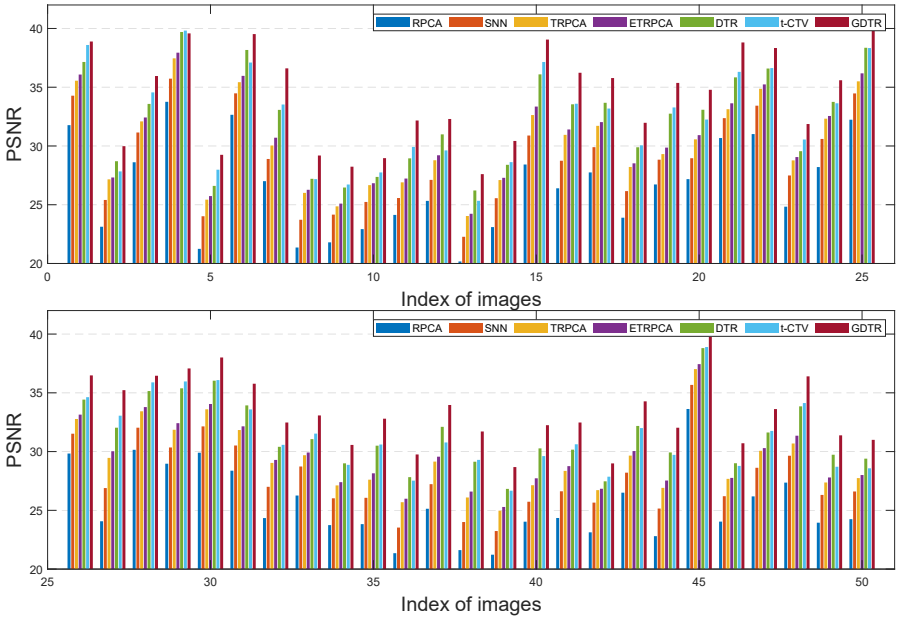



Figure 1: PSNR comparison of different methods on 50 BSD images for color image recovery. The first and second rows correspond to the first and last 25 images, respectively. The results are best viewed by zooming in on a computer screen.

which purely employs low-rankness prior. This is due to that TRPCA is based on the original TNN and ignores the tube sparse structure of the singular value tensor. Besides, DTR simultaneously takes into account the prior information of the tube sparse noise tensor and thus achieves better recovery results.

- Secondly, DTR retains more color information compared with TRPCA and ETRPCA. Although ETRPCA also adopts the weighting strategy, it utilizes a fixed weight vector for different singular values. Our proposed method DTR assigns adaptive weights for different singular tubes and the larger singular tubes that often correspond to color information are shrunk less, which offers explanation for the better recovery performance of DTR.

- Thirdly, the performance of GDTR is the most outstanding among all methods. t-CTV has been validated to be highly effective between models which simultaneously consider the low-rankness and smoothness prior [4]. However, GDTR is more superior than t-CTV in the color image recovery task, which demonstrates the effectiveness of the extension of DTR with smoothness.

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